

AI / Bio-AI 개발 프로세스 벤치마킹 보고서

OpenAI, Anthropic, Google DeepMind, DeepSeek, LLM/Agent/RAG/AI-Ops/Bio-AI 기업 공개 자료 기반

목적: Asperitas 개발 과정과 비교 분석하고, 벤치마킹 가능한 개발 원칙을 DB화하여 독자적 Biological Intelligence Factory Workflow로 고착화하기 위한 내부 작업 문서

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Executive bottom line: 이 문서의 결론은 단순하다. 최상위 AI 회사와 Bio-AI 회사들의 공통 개발 원리는 “모델”이 아니라 “검증 가능한 루프”다. 소스/데이터 -> 스키마/도구 -> 에이전트/워크플로우 -> 평가/관측 -> 인간 승인 -> 실험/제품 피드백으로 이어지는 반복 시스템을 먼저 구축해야 한다.

0. 사용 범위와 증거 경계

본 문서는 공개 웹 자료, 공식 문서, 공식 GitHub, 논문, 신뢰 가능한 산업/기관 자료를 기반으로 한 벤치마킹 보고서다. 각 회사의 내부 비공개 개발 프로세스 전체를 확인한 것이 아니며, 공개된 오픈소스/문서/논문에서 관찰 가능한 개발 원리만 구조화했다. DB에 넣을 때는 반드시 source registry, license/terms review, provenance, ingestion_status, confidence를 분리해야 한다.

- 공식 문서/공식 GitHub/논문은 우선 사용했다. 블로그·뉴스는 보조 신호로만 사용했다.
- “오픈소스”는 회사 전체 기술이 공개됐다는 뜻이 아니다. 대다수 Bio-AI 회사는 핵심 데이터와 모델이 비공개다.
- Benchmark는 Asperitas 사실을 대체하지 않는다. 외부 자료는 operating analogy와 workflow primitive로만 흡수한다.
- 생물학/컴플라이언스/투자자-facing claim은 반드시 근거/검증상태/승인상태를 분리한다.

1. Executive Bottom Line

가장 중요한 패턴은 4개다. 첫째, OpenAI/Anthropic류 LLM 회사는 “모델 성능”보다 “행동 명세, 평가, 도구 권한, 관측 가능성”을 운영체계화한다. 둘째, LangChain/LlamaIndex/CrewAI류 에이전트 회사는 자유로운 에이전트보다 상태 그래프, step/event workflow, guardrail, trace를 강조한다. 셋째, Hugging Face/Baseten/Modal/Databricks류 개발 운영 회사는 모델을 제품화하려면 패키징, 서빙, 비용, latency, rollback, monitoring이 필요하다는 것을 보여준다. 넷째, Recursion/Ginkgo/Isomorphic/Owkin/GenerateBio/Xaira류 Bio-AI 회사는 “AI prediction”보다 “proprietary data + metadata + wet-lab/assay validation + DBTL feedback loop”를 실제 moat로 만든다.

Asperitas에 대한 결론: 지금 단계의 최적 방향은 자율 실험실이 아니라 Benchmark Source Registry + AOS-aligned RAG/Agent + Eval/Trace/Compliance Gate + Bio Opportunity/DBTL Queue다. 이후 PTMC/VITRAYA 등 제한된 프로젝트에서 인간승인형 DBTL loop를 붙이는 것이 맞다.

2. 공통 개발 프로세스: 경쟁사들이 반복하는 “진짜 루프”

Pattern	Seen in	Operational meaning for Asperitas
Spec -> Eval -> Deployment loop	OpenAI, Anthropic, W&B, MLflow	Every behavior rule becomes testable. Build evals before expanding autonomy.
Constitution / policy layer	Anthropic, OpenAI Model Spec, Asperitas AOS	AOS should be compiled into validators, not left as prose.
Typed tool/function contracts	OpenAI Agents, Gemini, MCP, Ginkgo CFPS	Every tool call must have schema, permission, input validation, and human approval mode.
Source-grounded retrieval	LangChain, LlamaIndex, Perplexity, Glean	Separate internal truth, external current intelligence, and unverified benchmark sources.
Traceability / observability	W&B Weave, CrewAI, MLflow	Run logs, tool logs, decision logs, cost, latency, source citations, and error taxonomy are mandatory.
Biological data flywheel	Recursion, Ginkgo, Isomorphic, Xaira	Validated proprietary data beats generic model access. Build data acquisition and metadata systems now.
Closed-loop DBTL	Ginkgo/OpenAI, Recursion, Xaira	Hypothesis -> experiment spec -> execution -> assay -> learning update. Human-in-loop until safety/compliance proves stable.
Knowledge graph foundation	BenevolentAI, Glean, biomedical KG-RAG literature	Biology needs entity/evidence relations, not only vector chunks.
Privacy-preserving collaboration	Owkin/Substra	Partner datasets may remain distributed; compute and evidence can move instead of raw data.
Protein/structure generation as module	AlphaFold/Isomorphic, Chroma, AtomNet	Use prediction/generation as decision support; validation/IP/compliance gates determine value.

3. LLM / Foundation Model 회사별 개발 프로세스 벤치마킹

이 섹션은 LLM 회사들이 공개한 오픈소스, API 문서, 모델/에이전트 개발 자료에서 추출한 개발 강조점이다. Asperitas는 여기를 “모델을 직접 학습하는 방법”보다 “AI 시스템을 검증하고 운영하는 방식”으로 흡수해야 한다.

Company / stack	Public/open artifact	Observed development process emphasis	Asperitas benchmark takeaway
OpenAI	Evals, Model Spec evals, Agents SDK, structured-output eval cookbook [S01-S04]	Behavior spec -> eval harness -> typed agent outputs -> approval interruption. OpenAI's public tooling pushes teams to encode success criteria as tests rather than relying on vibes.	Adopt: every AOS/agent behavior rule becomes an eval. No strategic claim, RAG answer, or DBTL recommendation ships without schema, citation, and regression test.
Anthropic	Claude Constitution, Agent SDK, MCP [S05-S07]	Principle-governed model behavior plus production agent primitives: file tools, command execution, context management, permissions, hooks, and external tool/data protocol.	Adopt: AOS constitution + deterministic hooks. Use MCP-like integration but require strict tool permissions, audit log, allowlists, and prompt-injection tests.
Google / Gemini	Gemini API Cookbook and function calling docs [S08-S09]	Multimodal/action model workflow: practical cookbook, function declarations, external APIs, and agentic action patterns.	Adopt: Gemini-style function contracts for every internal tool: search_source, read_source, check_compliance, create_decision_log, queue_experiment.
Google DeepMind / AlphaFold	AlphaFold/Isomorphic public tech narrative [S10-S11]	Model alone is insufficient: structure prediction becomes valuable when integrated into target/disease-agnostic drug design workflows and experimental decision loops.	Adopt: do not stop at prediction. Store confidence, input provenance, assay requirement, validation status, and IP/compliance implications.
Meta AI / Llama	Llama Cookbook [S14]	Open model ecosystem: inference, fine-tuning, RAG, multimodal recipes, community deployment patterns.	Use as local/private model option for low-cost internal tasks and eval baseline, not as sole intelligence layer.
Mistral AI	Platform, agents, evals, judges, guardrails, deployment portability [S15]	Enterprise AI stack emphasizes custom agents, observability, registry, guardrails, and portability from edge to cloud.	Benchmark for internal/external deployment separation: local dev, private inference, cloud fallback, full observability.
Cohere	Aya multilingual model and eval research [S16]	Data curation and multilingual evaluation are treated as first-class infrastructure, not afterthoughts.	Useful for Korean/English/Chinese bio-intelligence. Build multilingual golden sets and translation-risk tests.
DeepSeek	DeepSeek-V3/R1 repos and reports [S12-S13]	Efficiency-first model engineering: MoE, MLA, cost-aware training, RL/reasoning release strategy.	Benchmark mindset: optimize cost per correct decision, not largest model. Use small specialized models after RAG/evals prove the task.
Qwen / Alibaba	Open model family and cookbook ecosystem (see source registry for active repo verification)	Broad open-weight strategy, multilingual/multimodal releases, ecosystem adoption.	Treat as deployable open baseline for internal/private workflows after security and license review.
Zhipu / Z.ai	GLM open-source hybrid reasoning repo [S17]	Hybrid reasoning modes: thinking for complex/tool tasks, non-thinking for fast responses.	Implement dual mode: fast answer for low-risk retrieval; reasoning+approval for bio/compliance/IP decisions.
xAI	Grok open releases and weights (source-map item; verify current license before ingestion)	Open-weight signaling with frontier-scale MoE framing, but limited process transparency compared with eval-heavy providers.	Use mainly as external benchmark signal, not as process template unless documentation/evals are stronger.

4. Agent / RAG / AI 개발운영 기업 벤치마킹

에이전트 생태계의 핵심은 “여러 에이전트가 알아서 한다”가 아니다. production-grade agent는 입력/출력 schema, 상태 관리, 도구 권한, 관측, 평가, rollback, human approval의 조합이다.

Company / stack	Public/open artifact	Observed development process emphasis	Asperitas benchmark takeaway
LangChain / LangGraph / LangSmith	LangGraph, LangSmith ecosystem [S18-S19]	Stateful graph orchestration, long-running agents, tracing, debugging, evals, and deployment management.	Use graph/state machine for Asperitas agents; no free-form autonomous loop for compliance-sensitive tasks.
LlamaIndex	LlamaIndex OSS, Workflows [S20-S21]	Data ingestion, parsing, indexing, agentic applications; event-driven async step workflows.	Best fit for source ingestion/RAG layer. Build source registry -> parser -> chunk -> index -> retrieval -> citation checker.

Company / stack	Public/open artifact	Observed development process emphasis	Asperitas benchmark takeaway
CrewAI	CrewAI docs/GitHub [S22]	Role-based crews, flows, guardrails, memory, knowledge, observability; deterministic workflow emphasis.	Use multi-agent roles only after core RAG/evals exist. Each agent must have single responsibility and explicit stop rules.
Weights & Biases	Weave [S23-S24]	Trace LLM inputs/outputs/tool calls, debug regressions, apples-to-apples evals, LLM judges/custom scorers.	Create internal trace/eval layer before expanding agents. Every answer should be replayable and scored.
Hugging Face	Transformers, TRL, PEFT [S25-S26]	Open model hub + post-training + parameter-efficient adaptation; lower barrier to custom models.	After source corpus/evals stabilize, train small adapters for Asperitas style/classification, not before.
Baseten	Truss [S27]	Package model into deployable container with config, hardware, runtime/dependency control.	Adopt model packaging discipline: config.yaml, reproducible runtime, rollback, endpoint tests.
Modal	Serverless GPU/inference/training/sandboxes [S28]	Elastic compute, low-latency inference, isolated sandboxes for AI-generated code, batch scale-out.	Use for bursty eval/inference jobs after cost guardrails. Avoid always-on GPU burn in Phase 0.
Databricks / MLflow	MLflow + Model Serving [S29-S30]	Debug/evaluate/monitor/optimize AI applications; serve models through governed REST endpoints.	Benchmark for production-grade governance: registry, serving endpoint, monitoring, quality/cost metrics.
Perplexity	Search API, Sonar API [S31-S32]	Web-grounded search and AI responses with real-time ranked results, streaming and source controls.	Use as external intelligence layer for market/regulatory/news; separate from internal source-of-truth.
Glean	Enterprise search/knowledge graph [S33]	Enterprise data graph, people/content/activity signals, governance-aware search and assistant layer.	Model Asperitas KB as enterprise knowledge graph: documents, people, species, protocols, decisions, experiments.

5. Bio-AI / AI Drug Discovery / Generative Biology 기업 벤치마킹

Bio-AI 회사들의 공통점은 “생물학적 데이터의 반복 생산과 검증”이다. 공개된 오픈소스는 일부 도구나 데이터셋에 그치지만, 개발 철학은 명확하다: 데이터 생성/정규화/표현학습/예측/실험/검증/제품화가 하나의 flywheel로 연결되어야 한다.

Company / stack	Public/open artifact	Observed development process emphasis	Asperitas benchmark takeaway
Recursion	RxRx datasets, RxRx3, Recursion OS narrative [S34-S35, S51]	Industrial phenomics flywheel: perturbation data -> embeddings/models -> drug discovery decisions; public datasets are only a small window into proprietary scale.	Critical lesson: proprietary biological data and batch-corrected phenotypes are moat. Build Asperitas phenotype/omics/compound registry early.
Ginkgo + OpenAI	CFPS autonomous lab paper and GitHub interface [S36-S37]	LLM + cloud lab + schema validation + automation platform + iterative experiment design/execution/data analysis. Reported 40% specific cost reduction and 27% titer increase in CFPS in the public paper.	This is the strongest DBTL benchmark. Copy the architecture pattern, not the exact biology: schema-validated experiment specs, human approval, automated data capture, iteration logs.
Isomorphic Labs	Drug design engine and AlphaFold 3 tech pages [S10-S11]	Biomolecular structure prediction is extended into an end-to-end, target/disease-agnostic drug design engine.	Prediction must be embedded in workflow: target hypothesis, candidate design, assay, clinical translatability, IP, and go/no-go.
Insilico Medicine	DORA open-source announcement and GitHub [S38-S39]	Agentic research-document workflow with agile agents, engineered prompts, references, workflows, and proprietary databases.	Useful for Literature/Patent/Grant Agent. Keep citation/relevance checks; do not let document generation masquerade as validated biology.
Exscientia	UKRI case study [S47]	AI-designed drug candidates with experienced drug hunter feedback; design and experimental validation are fused.	Benchmark: AI proposes, medicinal chemistry and assays dispose. Build wet-lab and domain review gates.
Atomwise	AtomNet paper and Scientific Reports study [S45-S46]	Structure-based CNN virtual screening; large-scale virtual HTS can substitute some early HTS functions when validated.	For Asperitas: use virtual screening only as prioritization. Wet-lab validation and assay reproducibility remain mandatory.

Company / stack	Public/open artifact	Observed development process emphasis	Asperitas benchmark takeaway
BenevolentAI	Knowledge graph/data-foundation material [S44]	Knowledge graph + proprietary ontologies + tooling on top for target/hypothesis generation.	Build KG not as ornament but as discovery engine: entity resolution, evidence paths, confidence, contradictory evidence.
Owkin	Substra federated learning [S42-S43]	Privacy-preserving collaborative model training on distributed medical datasets; production FL environments.	For biodiversity/hospital/partner data: do not centralize everything by default. Federated or partner-side compute can unlock regulated data.
Generate:Biomedicines	Chroma GitHub and Nature paper [S40-S41]	Programmable protein generation with composable building blocks, all-atom structure/sequence modeling and conditioning.	Asperitas should treat generative protein design as a module attached to proprietary organism/protein data and DBTL validation.
Formation Bio	Sanofi/OpenAI collaboration and Muse [S48-S49]	AI-native pharma operations across drug-development lifecycle; patient recruitment and development software.	Benchmark non-discovery bottlenecks too: trial design, recruitment, document ops, asset diligence, and portfolio execution.
Xaira Therapeutics	Official site and public launch narrative [S50]	Integrated AI biotech combining ML research, expansive data generation, and therapeutic product development.	Closest strategic analogy to a full-stack AI-bio company. Asperitas version must swap therapeutic-only focus for biodiversity-derived biological infrastructure.

6. Asperitas에 바로 적용할 Workflow v0.1

6.1 Benchmark-to-DB ingestion workflow

- 1) Source discovery: 회사명, 소스명, URL, source_type, official 여부, license/terms 상태를 먼저 registry에 기록한다.
- 2) Evidence classification: official docs, GitHub repo, paper, blog, news, inference를 분리한다.
- 3) Workflow primitive extraction: eval, agent loop, data flywheel, tool contract, observability, DBTL loop 등 재사용 가능한 패턴만 추출한다.
- 4) Asperitas fit scoring: Scalability, Moat, Biosafety/Compliance, Capital Efficiency, Learning Velocity 기준으로 1-5점 평가한다.
- 5) AOS integration: 통과한 패턴만 AGENTS.md, evals, schemas, runbooks, source policy에 반영한다.
- 6) Regression: 반영 전후 golden question/eval 통과 여부를 기록한다.

6.2 Internal agent build sequence

Phase	Objective	Build now / later	Benchmark sources
Phase 0 - Internal credibility memory	Create source-grounded operating memory, benchmark DB, citations, evals, compliance gates.	Build now: source registry, metadata, parsers, RAG, agent harness, trace/eval, report generator, decision log. Do not build autonomous wet-lab execution yet.	OpenAI evals, LlamaIndex, LangGraph, W&B, Glean
Phase 1 - Human-in-loop bio intelligence	Turn documents and datasets into structured bio opportunity cards and DBTL queues.	Build next: species/protein/pathway cards, literature/patent agent, compliance gatekeeper, experiment registry, assay-result schema.	Recursion, BenevolentAI, Owkin, GenerateBio
Phase 2 - Semi-automated DBTL	Use AI to recommend next experiments and capture results, with human approval and lab integration.	Build after data maturity: protocol registry, ELN/LIMS-ready schema, image/phenotype analysis, human-in-loop active learning.	Ginkgo/OpenAI autonomous lab, Recursion, Xaira
Phase 3 - Biological Intelligence Factory	Repeatable data/model/experiment/IP/product flywheel.	Build only after validated proprietary datasets and compliant execution infrastructure exist.	Isomorphic, Recursion, Xaira, Formation Bio

6.3 Recommended software modules

- source_registry: 외부 benchmark source와 내부 Asperitas source를 같은 registry에서 관리하되 priority/disclosure는 분리.
- benchmark_parser: GitHub README, API docs, papers, PDF, company pages를 markdown + metadata로 추출.
- pattern_extractor: “개발 프로세스/강조점/오픈소스 artifact/Asperitas 적용점”을 JSON으로 추출.
- comparison_engine: 현재 AOS/Codex roadmap 대비 gap과 action item 생성.
- eval_harness: citation accuracy, unsupported claim, compliance escalation, phase discipline, Korean COO style 검증.
- workflow_compiler: 통과한 benchmark pattern만 runbook, Codex prompt, AGENTS.md patch proposal로 변환.

7. 최소 DB Schema 제안

아래 스키마는 “나중에 학습시킬 예정”이라는 목표를 구현 가능한 DB 구조로 바꾸는 최소 단위다. 핵심은 원문을 무작정 넣는 것이 아니라, 출처/라이선스/검증상태/적용가능성/위험을 함께 넣는 것이다.

Table	Core fields	Purpose
benchmark_company	company_id, name, domain, category, source_priority, confidence, last_verified	Company-level benchmark registry.
benchmark_artifact	artifact_id, company_id, artifact_type, url, license_status, open_source_status, checksum, ingestion_status	Track repos, docs, papers, APIs, cookbooks.
process_pattern	pattern_id, name, source_artifacts, workflow_step, evidence_type, applicability, risk	Extract reusable workflow primitives from competitors.
asperitas_gap	gap_id, benchmark_pattern, current_state, required_change, owner, deadline, score	Compare benchmark to current Asperitas roadmap.
eval_case	eval_id, target_agent, prompt, expected_behavior, source_requirement, compliance_trigger, pass_rule	Regression tests for agent quality and safety.
decision_log	decision_id, context, sources_used, chosen_path, rejected_alternatives, risk, next_action, review_date	Preserve strategic memory and audit trail.
dbtl_event	event_id, hypothesis, design, protocol_version, approval_status, assay_result, validation_status, learning_update	Connect biological experiments to model learning.

7.1 JSONL record example

```
{
  "artifact_id": "BENCH-OPENAI-EVALS-001",
  "company": "OpenAI",
  "category": "LLM_eval_ops",
  "source_type": "official_github",
  "url": "https://github.com/openai/evals",
  "license_status": "verify_before_ingestion",
  "process_pattern": "behavior_spec_to_eval_harness",
  "asperitas_application": "Convert AOS rules into golden evals and regression tests",
  "evidence_label": "document_supported",
  "confidence": "high",
  "ingestion_status": "source_mapped_not_ingested",
  "risk": "do not store private eval data in public repos",
  "next_action": "Create eval_cases for citation accuracy, compliance escalation, phase discipline"
}
```

8. 우선순위 Action Plan

Priority	Action	Definition of done	Why now
P0	Benchmark Source Registry 생성	50개 이상 source row: source_type, URL, license_status, confidence, ingestion_status 입력	자료를 “금음”과 “DB화”를 분리해야 환각/저작권/출처 봉괴를 막는다.
P0	Golden Eval 20개 작성	citation accuracy, unsupported claim, compliance, phase discipline, Korean COO style 테스트 통과	최고 기업들은 eval로 운영한다. 에이전트 성능 향상은 eval 없이는 감으로 전략한다.
P0	Benchmark Pattern JSON 추출기	각 source에서 process_pattern, applicability, risk, action_item 추출	PDF 보고서가 DB 입력으로 이어져야 한다.
P1	AOS-Codex Roadmap Gap Matrix	현재 repo roadmap과 benchmark pattern 간 gap/owner/deadline 작성	벤치마킹이 실제 개발 순서 수정으로 연결된다.
P1	Human-in-loop DBTL schema	hypothesis, design, approval, protocol, assay, result, validation fields 정의	Bio-AI moat는 실험 데이터와 검증 label에서 생긴다.
P2	Bio opportunity cards MVP	species/protein/pathway/literature/compliance/IP 카드 자동 생성	COO/연구팀이 매일 쓸 수 있는 내부 intelligence product로 전환된다.

9. Devil’s Advocate: 벤치마킹할 때 반드시 피해야 할 착각

- 오픈소스 repo를 모았다고 그 회사의 moat를 복제한 것이 아니다. Recursion/Isomorphic/Xaira의 핵심은 공개 repo가 아니라 데이터, 실험, 팀, 파트너, capital, validation loop다.
- 에이전트를 많이 나누면 성능이 올라가는 것이 아니다. source grounding, eval, trace, tool permission이 없으면 다중 에이전트는 환각을 여러 명이 나눠서 하는 구조가 된다.
- Bio-AI에서 prediction은 validation이 아니다. 구조 예측, virtual screening, protein generation은 prioritization signal이지 wet-lab proof가 아니다.
- 웹 크롤링은 DB화가 아니다. license/terms, provenance, redaction, checksum, metadata, ingestion logs가 있어야 DB화다.
- Phase 0에서 autonomous wet-lab을 만들면 안 된다. 먼저 source registry, schema, compliance gate, eval, decision log를 완성해야 한다.

10. Source Registry: 이번 보고서에 사용한 핵심 공개 자료

DB에 실제로 넣을 때는 아래 각 row를 benchmark_artifact 테이블로 이동시키고, license_status와 ingestion_status를 별도로 관리한다. URL은 공개 접근 가능한 위치를 가리키며, 이용약관/라이선스 검토 전에는 원문 대량 저장을 하지 않는다.

ID	Source	Organization	Use	URL
S01	OpenAI Evals GitHub	OpenAI	Eval framework for LLMs/systems; private/custom eval patterns	https://github.com/openai/evals
S02	OpenAI structured output eval cookbook	OpenAI	Structured output evaluation example with Evals API	https://developers.openai.com/cookbook/examples/evaluation/use-cases/structured-outputs-evaluation
S03	OpenAI Model Spec / model_spec_evals	OpenAI	Desired model behavior and evaluation harness	https://github.com/openai/model_spec_evals
S04	OpenAI Agents SDK results	OpenAI	Typed final outputs and approval interruption patterns	https://openai.github.io/openai-agents-python/results/
S05	Claude Agent SDK overview	Anthropic	Production agent loop, tools, context management	https://code.claude.com/docs/en/agent-sdk/overview
S06	Claude Constitution	Anthropic	Constitutional behavior framework	https://www.anthropic.com/constitution
S07	MCP engineering article	Anthropic	Open standard for connecting AI agents to external systems	https://www.anthropic.com/engineering/code-execution-with-mcp
S08	Gemini API Cookbook	Google	Structured learning path and practical API examples	https://github.com/google-gemini/cookbook
S09	Gemini function calling cookbook	Google	Function calling with REST / notebooks	https://github.com/google-gemini/cookbook/blob/main/quickstarts/Function_calling.ipynb

ID	Source	Organization	Use	URL
S10	AlphaFold 3 / Isomorphic Labs article	Isomorphic Labs	Drug design engine beyond structure prediction	https://www.isomorphiclabs.com/articles/the-isomorphic-labs-drug-design-engine-unlocks-a-new-frontier
S11	Isomorphic Labs Our Tech	Isomorphic Labs	AlphaFold 3 and end-to-end drug design engine framing	https://www.isomorphiclabs.com/our-tech
S12	DeepSeek-V3 GitHub	DeepSeek	MoE, MLA, DeepSeekMoE, multi-token prediction	https://github.com/deepseek-ai/deepseek-v3
S13	DeepSeek-R1 GitHub	DeepSeek	R1/R1-Zero derived from V3-base; open model downloads	https://github.com/deepseek-ai/deepseek-r1
S14	Meta Llama Cookbook	Meta AI	Inference, fine-tuning and end-to-end recipes	https://github.com/meta-llama/llama-cookbook
S15	Mistral AI platform page	Mistral AI	Enterprise AI platform: agents, evals, guardrails, deployment portability	https://mistral.ai/
S16	Cohere Aya paper	Cohere For AI	Multilingual instruction tuning/evaluation across 99+ languages	https://arxiv.org/abs/2402.07827
S17	Z.ai / GLM open-source repo	Z.ai	Hybrid reasoning modes and open model releases	https://github.com/zai-org/GLM-4.5
S18	LangGraph GitHub	LangChain	Long-running, stateful agent orchestration	https://github.com/langchain-ai/langgraph
S19	LangSmith / LangGraph docs via LangChain	LangChain	Agent orchestration, debug, scale, eval ecosystem	https://www.langchain.com/langgraph
S20	LlamaIndex GitHub	LlamaIndex	Agentic applications, parsing, extraction, indexing	https://github.com/run-llama/llama_index
S21	LlamaIndex Workflows	LlamaIndex	Event-driven, async, step-based workflows	https://github.com/run-llama/llama-agents
S22	CrewAI docs	CrewAI	Multi-agent systems with guardrails, memory, knowledge, observability	https://docs.crewai.com/
S23	W&B Weave GitHub	Weights & Biases	Trace/debug LLM IO and rigorous evaluations	https://github.com/wandb/weave
S24	W&B Weave docs	Weights & Biases	Observability and evaluation with LLM judges/custom scorers	https://docs.wandb.ai/weave
S25	Hugging Face TRL	Hugging Face	SFT/RLHF/GRPO-style post-training workflows	https://github.com/huggingface/trl
S26	Hugging Face PEFT	Hugging Face	Parameter-efficient fine-tuning to reduce compute/storage	https://github.com/huggingface/peft
S27	Baseten Truss	Baseten	Open-source packaging/deployment for model serving	https://github.com/basetenlabs/truss
S28	Modal docs	Modal	Serverless compute/inference/training/sandboxes	https://modal.com/docs/guide
S29	MLflow	Databricks/MLflow	Open-source AI engineering platform for agents, LLMs, ML	https://mlflow.org/
S30	Databricks Model Serving	Databricks	Unified REST API for real-time/batch model serving	https://docs.databricks.com/aws/en/machine-learning/model-serving/
S31	Perplexity API Platform	Perplexity	Real-time web search and grounded AI API	https://www.perplexity.ai/api-platform
S32	Perplexity Sonar API	Perplexity	Web-grounded AI responses with tools/search options	https://docs.perplexity.ai/docs/sonar/quickstart
S33	Glean Work AI / Knowledge Graph	Glean	Enterprise knowledge graph/search/agent governance	https://www.glean.com/
S34	Recursion RxRx Datasets	Recursion	Public research datasets repository	https://github.com/recursionpharma/rxrx-datasets
S35	RxRx3 public map of biology	Recursion	Public subset of Recursion dataset, <1% of total	https://www.rxrx.ai/rxrx3

ID	Source	Organization	Use	URL
S36	OpenAI/Ginkgo autonomous lab PDF	OpenAI/Ginkgo	GPT-5 autonomous lab for CFPS optimization	https://cdn.openai.com/pdf/5a12a3bc-96b7-4e07-9386-db6ee5bb2ed9/using-a-gpt-5-driven-autonomous-lab-to-optimize-the-cost-and-titer-of-cell-free-protein-synthesis.pdf
S37	Ginkgo CFPS automation interface GitHub	Ginkgo Bioworks	Interface between GPT-5 and autonomous laboratory	https://github.com/ginkgobioworks/ginkgo-automation-cfps
S38	Insilico DORA news	Insilico Medicine	Agentic research document drafting with agents/prompts/workflows	https://insilico.com/news/kv5e80yc41-insilico-open-sources-dora-on-github-sup
S39	Insilico GitHub	Insilico Medicine	DORA and other public repos	https://github.com/insilicomedicine
S40	GenerateBio Chroma GitHub	Generate:Biomedicines	Programmable generative protein design model	https://github.com/generatebio/chroma
S41	Chroma Nature paper	Generate:Biomedicines	Programmable generative protein design	https://www.nature.com/articles/s41586-023-06728-8
S42	Owkin/Substra GitHub	Owkin / LF AI & Data	Open-source federated learning software	https://github.com/substra
S43	Substra docs	Owkin / LF AI & Data	Production federated learning environments	https://docs.substra.org/
S44	BenevolentAI data foundations	BenevolentAI	Knowledge graph enriched with data and products/tools on top	https://www.benevolent.com/news-and-media/blog-and-videos/building-data-foundations-accelerate-drug-discovery/
S45	AtomNet paper	Atomwise	Structure-based deep convolutional network for bioactivity prediction	https://arxiv.org/abs/1510.02855
S46	AtomNet Scientific Reports 2024	Atomwise	Virtual HTS campaign with AtomNet	https://www.nature.com/articles/s41598-024-54655-z
S47	Exscientia UKRI case study	Exscientia	AI-designed drug candidate pipeline	https://www.ukri.org/who-we-are/how-we-are-doing/research-outcomes-and-impact/bbsrc/exscientia-a-clinical-pipeline-for-ai-designed-drug-candidates/
S48	Sanofi-Formation Bio-OpenAI announcement	Formation Bio / Sanofi / OpenAI	AI-powered software across drug development lifecycle	https://www.sanofi.com/en/media-room/press-releases/2024/2024-05-21-05-30-00-2885244
S49	Formation Bio Muse	Formation Bio	AI tool for patient recruitment in drug development	https://www.formation.bio/press/introducing-muse
S50	Xaira Therapeutics official	Xaira Therapeutics	Predictive AI across target, therapeutic, patient selection	https://www.xaira.com/
S51	Recursion official technology page	Recursion	Recursion OS, data, ML models, BioHive-2	https://www.recursion.com/

Appendix A. 벤치마킹 흡수 규칙

- Rule 1 - Benchmark source is not company fact. 외부 회사 자료는 Asperitas의 실제 구축 상태를 증명하지 않는다.
- Rule 2 - Open-source status must be checked per artifact. GitHub repo가 있어도 모델/데이터/weights/API/라이선스 범위가 다르다.
- Rule 3 - Every claim gets evidence_label. fact, inference, speculation, verification_needed를 분리한다.
- Rule 4 - Compliance-first bio workflow. CITES, Nagoya, LMO, biosafety, biosecurity, IP, privacy는 후처리 항목이 아니라 설계 항목이다.
- Rule 5 - Phase discipline. Phase 0은 internal source-grounded decision system, Phase 1은 human-in-loop bio intelligence, Phase 2는 semi-automated DBTL, Phase 3는 full Biological Intelligence Factory다.

Appendix B. Codex에 줄 최소 명령어 초안

Build BENCHMARK_DB_V0 for Asperitas.

Goal: turn public AI/Bio-AI company development-process sources into source-grounded benchmark records and compare them against current Asperitas AOS/Codex roadmap.

Create: data/benchmark_sources.csv, data/benchmark_patterns.jsonl, app/benchmark_compare.py, evals/test_benchmark_grounding.py, docs/BENCHMARK_WORKFLOW.md.

Rules: no uncontrolled scraping; store URL/provenance/license_status/ingestion_status; separate official docs, GitHub, paper, news, inference; no claim becomes fact without source; no external benchmark overwrites internal Asperitas truth.

Output: gap matrix with action_item, owner_placeholder, priority, risk, phase, eval_required.

Tests: citation accuracy, source-type labeling, unsupported-claim refusal, compliance escalation, phase discipline.